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Off-Campus: Bachupally-Gandimaisamma Road, Bowrampet, Hyderabad, Telangana - 500 043.
Phone No: 7815926816, www.klh.edu.in

STUDENT CASE STUDY EXECUTION REPORT

Department	FED
Subject	DSA-2
Academic Year	I-III Semester(2025-26)
Faculty Name	Ms. Alluri Swathi
Student Name	N. Tejasri
Roll Number	2520030132
Branch	CSE
Course Outcome	6

Case Study Topic: Digital Voting & Election Analytics System using 0/1 Knapsack

Work Done Summary:

Implemented the 0/1 Knapsack Algorithm in Java for election resource optimization and secure voting management. Modeled polling stations as items with required resources (weight) and priority values (importance). The 0/1 Knapsack algorithm was used to select the optimal set of polling stations within the available election budget or resource capacity. Implemented voter registration, authentication verification, ballot casting, vote counting, election analytics, and fraud detection along with optimal resource allocation. Demonstrated efficient selection of polling stations to maximize election coverage while minimizing resource usage. Compared **0/1** Knapsack with a Greedy approach – 0/1 Knapsack provides an optimal solution for resource allocation using Dynamic Programming, making it suitable for secure election management.

Implementation Details :

1. Created a PollingStation class with fields: stationId, stationName, resourceRequired, and priorityValueStored candidate records in an array.
2. Stored polling station records in an array.
3. Implemented the 0/1 Knapsack Dynamic Programming algorithm.
4. Used resourceRequired as the weight and priorityValue as the value for each polling station.
5. Calculated the maximum achievable priority within the available resource capacity
6. Selected the optimal combination of polling stations for election deployment.
7. Performed fraud detection by identifying duplicate or suspicious polling station entries before optimization.
8. The main method inserted polling station data and demonstrated 0/1 Knapsack-based election resource optimization.

Result : Screenshots/Output

```
PS D:\JavaPrograms> javac SecureVoteKnapsack.java
PS D:\JavaPrograms> java SecureVoteKnapsack
===== SecureVote Digital Voting System =====

Registered Voters:
V101
V102
V103
V104
V105

Enter Voter ID: V101
Authentication Successful

Vote Count
Candidate A : 120
Candidate B : 150
Candidate C : 95

Fraud Detection
No duplicate votes detected.

Election Resource Optimization using 0/1 Knapsack

Maximum Priority Value = 47

Selected Polling Stations:
4  Station D  Resource=5  Priority=22
2  Station B  Resource=3  Priority=15
1  Station A  Resource=2  Priority=10
PS D:\JavaPrograms>
```

Time Complexity:

- Dynamic Programming Table Construction: **O(n × W)**
- Space Complexity: **O(n × W)**
- 0/1 Knapsack efficiently optimizes election resource allocation and polling station selection while maximizing election coverage within limited resources.

- **GitHub Link :** https://github.com/tejasri3123/Dsa-2

Student Signature	**Faculty Signature**